

Chapter Three: Theory of Consumer Behaviour

- ❖ **Consumer theory** is based on what people like, so it begins with something that we can't directly measure, but must infer.
- ❖ That is, **consumer theory** is based on the premise that we can infer what people like from the choices they make.
- ❖ **Consumer behaviour** can be best understood in **three steps**.
 - 1)**First**, by examining **consumer's preference**, we need a practical way to describe how people prefer one good to another.
 - 2)**Second**, we must take into account that consumers face **budget constraints** – they have limited incomes that restrict the quantities of goods they can buy.
 - 3)**Third**, we will put **consumer preference** and **budget constraint** together to determine consumer choice.

3.1 Consumer preferences

- ❖ A consumer makes **choices** by **comparing bundle of goods**.
- ❖ Given any two consumption bundles, the consumer either decides that one of the consumption **bundles is strictly better than the other**, or decides that she is **indifferent** between the two bundles.
- ❖ In order to tell whether one bundle is preferred to another, we see how the consumer behaves in choice situations involving two bundles.
- ❖ **If she always chooses X when Y is available**, then it is natural to say that this consumer prefers X to Y. We use the symbol $>$ to mean that one bundle is strictly preferred to another, so that **$X > Y$** should be interpreted as saying that the consumer **strictly prefers X to Y**, in the sense that she definitely wants the X-bundle rather than the Y-bundle.
- ❖ If the consumer is **indifferent between two bundles** of goods, we use the symbol $\sim$ and write **$X \sim Y$** .

- **Indifference** means that the consumer would be just as satisfied, according to her own preferences, consuming the bundle X as she would be consuming bundle Y .
- If the consumer **prefers or is indifferent between the two bundles** we say that she weakly prefers X to Y and write $X \succeq Y$.
- The relations of **strict preference**, **weak preference**, and **indifference** are **not independent concepts**; the relations are themselves related.
- **For example:**
 - ✓ if $X \succeq Y$ and $Y \succeq X$, we can conclude that $X \sim Y$. That is, if the consumer thinks that X is at least as good as Y and that Y is at least as good as X , then she must be **indifferent between the two** bundles of goods.
 - ✓ **Similarly**, if $X \succeq Y$ but we know that it is not the case that $X \sim Y$, we can conclude that $X \succ Y$. This just says that if the consumer thinks that X is at least as good as Y , and she is not indifferent between the two bundles, then she thinks that **X is strictly better than Y** .

3.2 The concept of utility

- ❖ Economists use the term **utility** to describe the **satisfaction** or **pleasure** derived from the consumption of a good or service.
- ❖ In other words, utility is the power of the product to satisfy human wants.
- ❖ Given any two consumption bundles X and Y, the consumer definitely wants the X-bundle than the Y-bundle if and only if the utility of X is better than the utility of Y.

Do you think that utility and usefulness are synonymous?

- ✓ **In defining utility**, it is important to bear in mind **the following points.**

1) **Utility** and **Usefulness** are not synonymous.

For example: Paintings by Picasso may be useless functionally but offer great utility to art lovers.

- ❖ Hence, **usefulness is product centric** whereas **utility is consumer centric.**

2) **Utility is subjective. Vary from person to person.** That means, the utility that two individuals derive from consuming the same level of a product may not be the same. **For example:** non-smokers do not derive any utility from cigarettes.

2) **Utility** can be **different at different places** and **time**.

For example: the utility that we get from drinking coffee early in the morning may be different from the utility we get during lunch time.

3.3 Approaches of measuring utility

✓ **Two major approaches** to measure utility: **cardinal** and **ordinal** approaches.

❖ The cardinalist school postulated that **utility can be measured objectively**.

❖ According to the **ordinalist** school, **utility is not measurable in cardinal numbers** rather the consumer can rank or order the utility he derives from different goods and services.

3.3.1 The cardinal utility theory

- ❖ According to the **cardinal utility** theory, utility is measurable by arbitrary unit of measurement called **utils** in the form of 1, 2, 3 etc.
For example: we may say that consumption of an orange gives Bilen 10 utils and a banana gives her 8 utils, and so on.
- ❖ From this, we can assert that **Bilen** gets more satisfaction from **orange** than from **banana**.

3.3.1.1 Assumptions of cardinal utility theory

- ❖ The cardinal approach is based on the following **major assumptions**.
 - 1) Rationality of consumers.** The **main objective** of the consumer is to maximize his/her satisfaction given his/her limited budget or income. Thus, in order to maximize his/her satisfaction, the consumer has to be rational.
 - 2) Utility is cardinally measurable.**
 - 3) Constant marginal utility of money.** A given unit of money deserves the same value at any time or place it is to be spent.

4) Diminishing marginal utility (DMU). The utility derived from each successive units of a commodity diminishes. In other words, the marginal utility of a commodity diminishes as the consumer acquires larger quantities of it.

5) The total utility of a basket of goods depends on the quantities of the individual commodities.

❖ If there are **n commodities** in the bundle with quantities **$X_1, X_2, X_3, \dots, X_n$** , the total utility is given by:

$$TU = f(X_1, X_2, X_3, \dots, X_n)$$

3.3.1.2 Total and marginal utility

A) Total Utility (TU)

- ❖ is the **total satisfaction** a consumer gets from consuming some specific quantities of a commodity **at a particular time**.
- ❖ As the consumer consumes **more of a good** per time period, his/her **total utility increases**.

- ❖ However, there is **a saturation point** for that commodity beyond which the consumer will not be capable of enjoying any greater satisfaction from it.

B) Marginal Utility (MU)

- ❖ is the **extra satisfaction** a consumer realizes from an additional unit of the product.
- ❖ In other words, marginal utility is the **change in total utility** that results from the consumption of one more unit of a product.
- ❖ Graphically, it is **the slope of total utility**.
- ❖ Mathematically, marginal utility is:

$$MU = \frac{\Delta TU}{\Delta Q}$$

- ❖ where, ΔTU is the change in total utility, and ΔQ is the change in the amount of product consumed.

Example: Lets consider a hypothetical example given in **Table 3.1** below.

Table 3.1: Total and marginal utility

Quantity	Total Utility(TU)	Marginal Utility(MU)
0	0	-
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2

❖ **Graphically**, the above data can be depicted as follows.

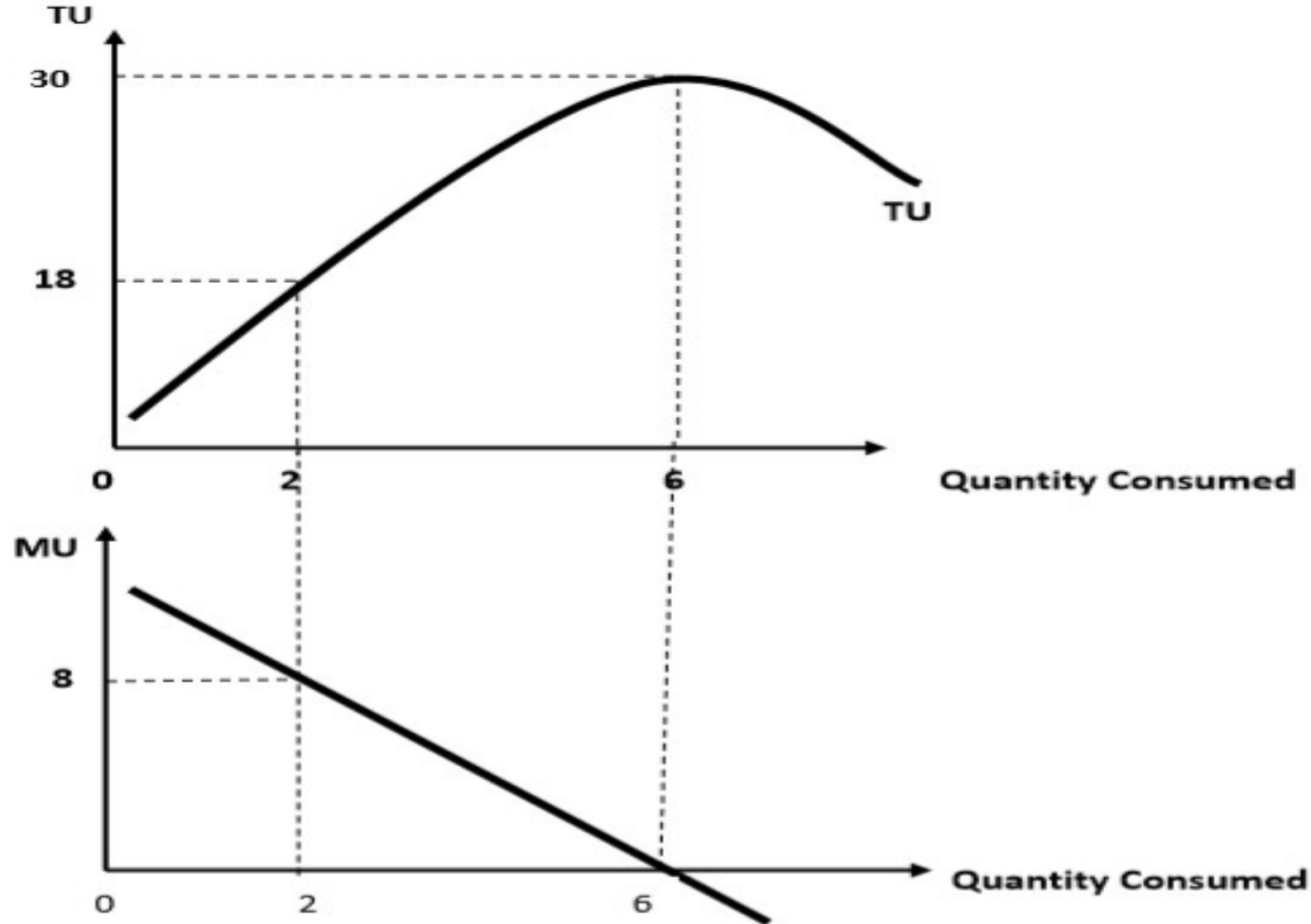


Figure 3.1: Total and marginal utility curves

- ❖ The **total utility first increases**, reaches the **maximum** (when the consumer consumes 6 units) and **then declines** as the quantity consumed increases.
- ❖ On the other hand, the **marginal utility continuously declines** (even becomes zero or negative) as quantity consumed increases.
- ❖ From the graph of **TU** and **MU**:
 - ✓ When **TU** is **increasing**, **MU** is **positive**.
 - ✓ When **TU** is **maximized**, **MU** is **zero**.
 - ✓ When **TU** is **decreasing**, **MU** is **negative**.

3.3.1.3 Law of diminishing marginal utility (LDMU)

Is the utility you get from consumption of the first orange the same as the second or the third orange?

- ❖ The **law of diminishing marginal utility** states that as the quantity consumed of a commodity increases per unit of time, **the utility derived from each successive unit decreases**, consumption of all other commodities remaining constant.
- ❖ In other words, the **extra satisfaction** that a consumer derives **declines** as he/she consumes more and more of the product **in a given period of time**.
- ❖ The **LDMU** is based on the following assumptions.
 - ✓ The **consumer** is **rational**
 - ✓ The consumer consumes **identical** or **homogenous product**. The commodity to be consumed should have **similar quality**, **color**, **design**, etc.
 - ✓ There is **no time gap/constant** in consumption of the good
 - ✓ The consumer **taste/preferences remain unchanged**

3.3.1.4 Equilibrium of a consumer

- ❖ The objective of a rational consumer is to maximize total utility.
- ❖ As long as the additional unit consumed brings a positive marginal utility, the consumer wants to consume more of the product because total utility increases.
- ❖ However, the consumer has income/budget constraint.

A) The case of one commodity

- ❖ The equilibrium condition of a consumer that consumes a single good **X** occurs when the marginal utility of X is equal to its market price.

$$MU_x = P_x$$

Proof

Given the **utility function**:

$$U = f(X)$$

- ❖ If the consumer buys commodity X, then his expenditure will be $P_x Q_x$.
- ❖ The consumer maximizes the **difference between his utility and expenditure.**

$$\max(U(Q_x) - P_x Q_x)$$

- ❖ The **necessary condition** for maximization is equating the derivative of a function to zero.
- ❖ **Thus,** $\frac{dU(Q_x)}{dQ_x} - \frac{dP_x Q_x}{dQ_x}$
- ❖ **Because ,** $\frac{dU(Q_x)}{dQ_x} = MU_x$

$$MU_x = P_x$$

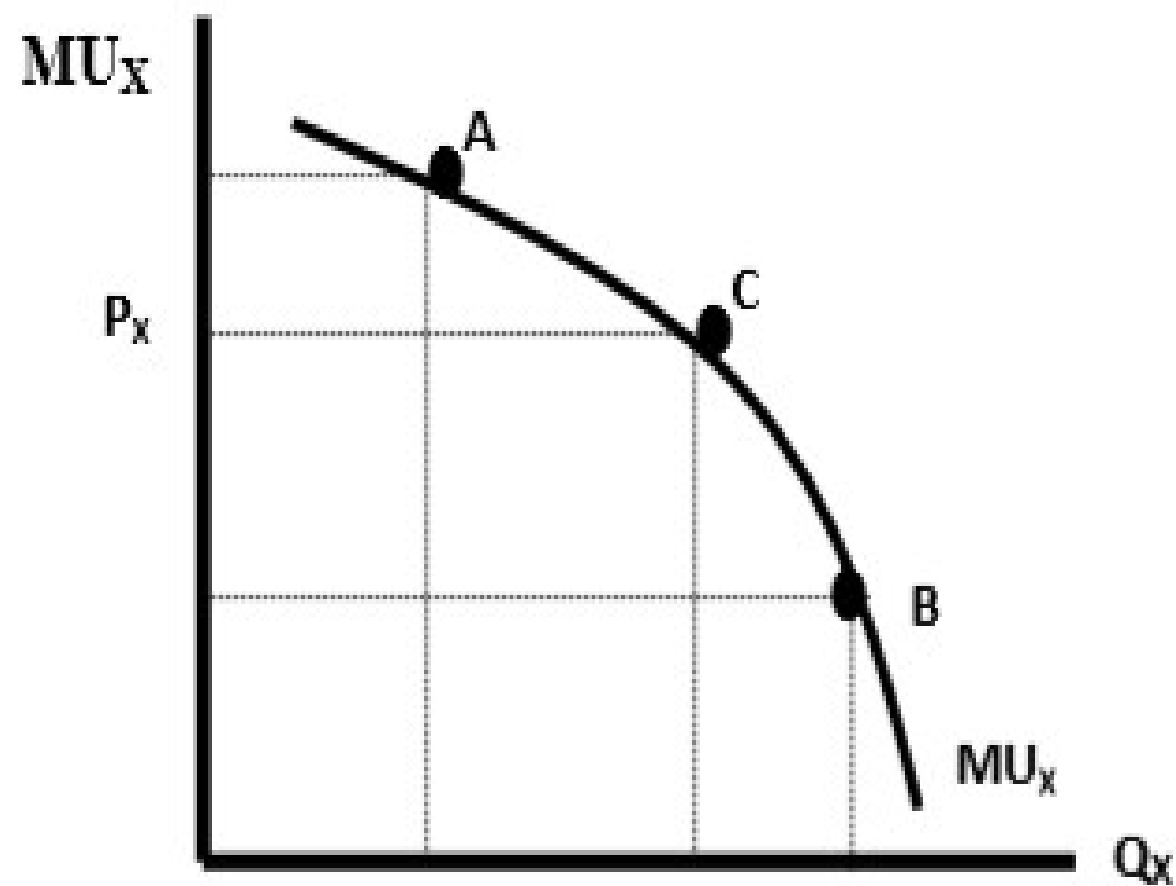


Figure 3.2: Equilibrium condition of consumer with only one commodity

- ❖ At any point **above point C** (like point A) where $MU_x > P_x$, it pays the consumer to consume more.
- ❖ When $MU_x < P_x$ (like point B), the **consumer should consume less of X**.
- ❖ At point C where $MU_x = P_x$ the consumer is at equilibrium.

B) The case of two or more commodities

- ❖ For the case of two or more goods, the consumer's equilibrium is achieved when the marginal utility per money spent is equal for each good purchased and his **money income** available for the purchase of the goods is exhausted.
- ❖ That is,
$$1) \frac{MU_x}{P_x} = \frac{MU_y}{P_y} = \dots = \frac{MU_n}{P_n}$$
$$2) P_x Q_x + P_y Q_y + \dots + P_n Q_n = M$$
- ❖ Where, **M** is the **income** of the consumer.

Example: Suppose Saron has $M=7$ two goods: **banana** and **bread**. The unit price of banana = 1 Birr and the unit price of a loaf of bread = 4 Birr.

Table 3.2: Utility schedule for two commodities

<i>Income = 7 Birr, Price of banana = 1 Birr, Price of bread = 4 Birr</i>							
<i>Banana</i>				<i>Bread</i>			
<i>Quantity</i>	<i>TU</i>	<i>MU</i>	<i>MU/P</i>	<i>Quantity</i>	<i>TU</i>	<i>MU</i>	<i>MU/P</i>
0	0	-	-	0	0	-	-
1	6	6	6	1	12	12	3
2	11	5	5	2	20	8	2
3	14	3	3	3	26	6	1.5
4	16	2	2	4	29	3	0.75
5	16	0	0	5	31	2	0.5
6	14	-2	-2	6	32	1	0.25

- ❖ The **equilibrium state** must satisfy the two conditions stated before.
- ❖ Accordingly, the equilibrium level of consumption for **Saron** will be:
Banana = 3 units, and
Bread = 1 loaf of bread.
- ❖ At this combination $P_x Q_x + P_y Q_y = M$
 That is $(1*3) + (4*1) = 7$
- ❖ Given this, the total utility that Saron derives from this combination can be given by:

$$TU = TU1 + TU2$$

$$TU = 14 + 12 \quad TU = 26$$
- ❖ Given her **fixed income** and the **price level of the two goods**, **no combination of the two goods will give her higher TU than this level of utility.**

Limitation of the cardinal approach

- 1) The assumption of cardinal utility is doubtful because **utility may not be quantified**. Utility cannot be measured absolutely (objectively).
- 2) The assumption of **constant MU of money** is **unrealistic** because as income increases, the **marginal utility of money changes**.

3.3.2 The ordinal utility theory

- ❖ In the ordinal utility approach, **it is not possible for consumers to express the utility of various commodities they consume in absolute terms, like 1 util, 2 utils, or 3 utils.**
- ❖ **However**, it is possible to express the **utility in relative terms**.
- ❖ The consumers **can rank commodities in the order of their preferences** as 1st, 2nd, 3rd and so on.
- ❖ **Therefore**, the consumer need not know in specific units the utility of various commodities to make his choice.

- ❖ It suffices for him to be able to **rank the various baskets of goods** according to the satisfaction that each bundle gives him.

3.3.2.1 Assumptions of ordinal utility theory

- ❖ The ordinal approach is based on the following assumptions.
 - 1)**Consumers are rational:** they maximize their satisfaction or utility given their income and market prices.
 - 2)**Utility is ordinal:** utility is not absolutely (cardinally) measurable. Consumers are required only to order or rank their preference for various bundles of commodities.
 - 3)**Diminishing marginal rate of substitution:** The marginal rate of substitution **is the rate at which a consumer is willing to substitute one commodity for another commodity** so that his total satisfaction remains the same.

4) The **total utility** of a consumer is measured by the amount (quantities) of all items he/she consumes from his/her consumption basket.

4) **Consumer's preferences are consistent.**

For example: if there are three goods in a given consumer's basket, say, X, Y, Z and if he prefers X to Y and Y to Z, then the consumer is expected to prefer X to Z. This property is known as **axioms of transitivity**.

❖ The **ordinal utility approach** is explained with the help of **indifference curves**.

❖ **Therefore**, the ordinal utility theory is also known as the **indifference curve approach**.

3.3.2.2 Indifference set, curve and map

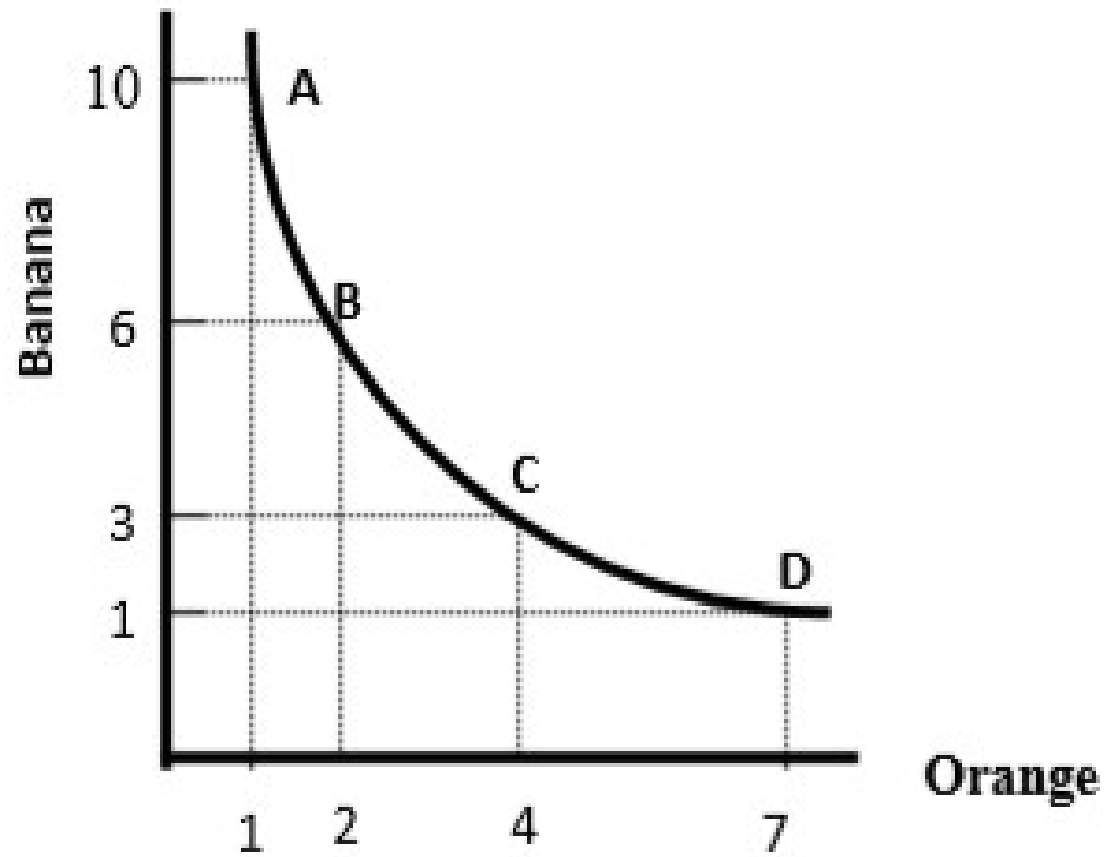
❖ **Indifference set/ schedule:** is a combination of goods for which the consumer **is indifferent**. It shows the **various combinations of goods** from which the consumer derives the **same level of satisfaction**.

Example: Consider a consumer who consumes two goods **X** and **Y**.

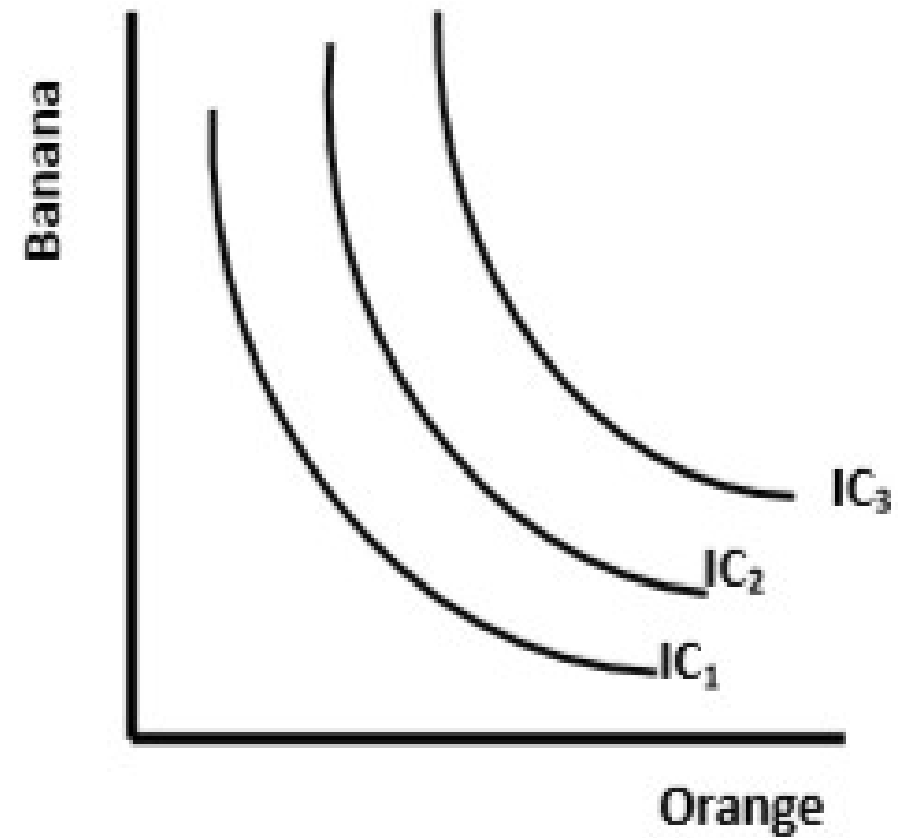
Table 3.3: Indifference schedule

Bundle(Combination	A	B	C	D
Orange	1	2	4	7
Banana	10	6	3	1

- ❖ In table 3.3 above, each combination of good **X** and **Y** gives the consumer equal level of total utility.
- ❖ Thus, the individual is indifferent whether he consumes combination A, B, C or D.
- ❖ **Indifference curve:** graphical expression of indifference set/schedule. It shows different combinations of two goods which yield the same utility (level of satisfaction) to the consumer.
- ❖ **Indifference map:** A set of indifference curves.



i) Indifference curve



ii) Indifference map

Figure 3.3: Indifference curve and indifference map

3.3.2.3 Properties of indifference curves

- 1) **Indifference curves** have **negative slope** (downward sloping to the right). because the consumption level of one commodity can be increased only by reducing the consumption level of the other commodity.
- 2) **Indifference curves** are **convex to the origin**.
 - ✓ The convexity of indifference curves is the reflection of **the diminishing marginal rate of substitution**. implies that the **commodities can substitute** one another but **are not perfect substitutes**.
- 3) **A higher indifference curve** is always preferred to a **lower one**.
- 3) **Indifference curves never cross each other** (cannot intersect).
 - ✓ The assumptions of consistency and transitivity will rule out the intersection of indifference curves.

Figure 3.4 shows the **violations of the assumptions of preferences** due to the intersection of indifference curves.

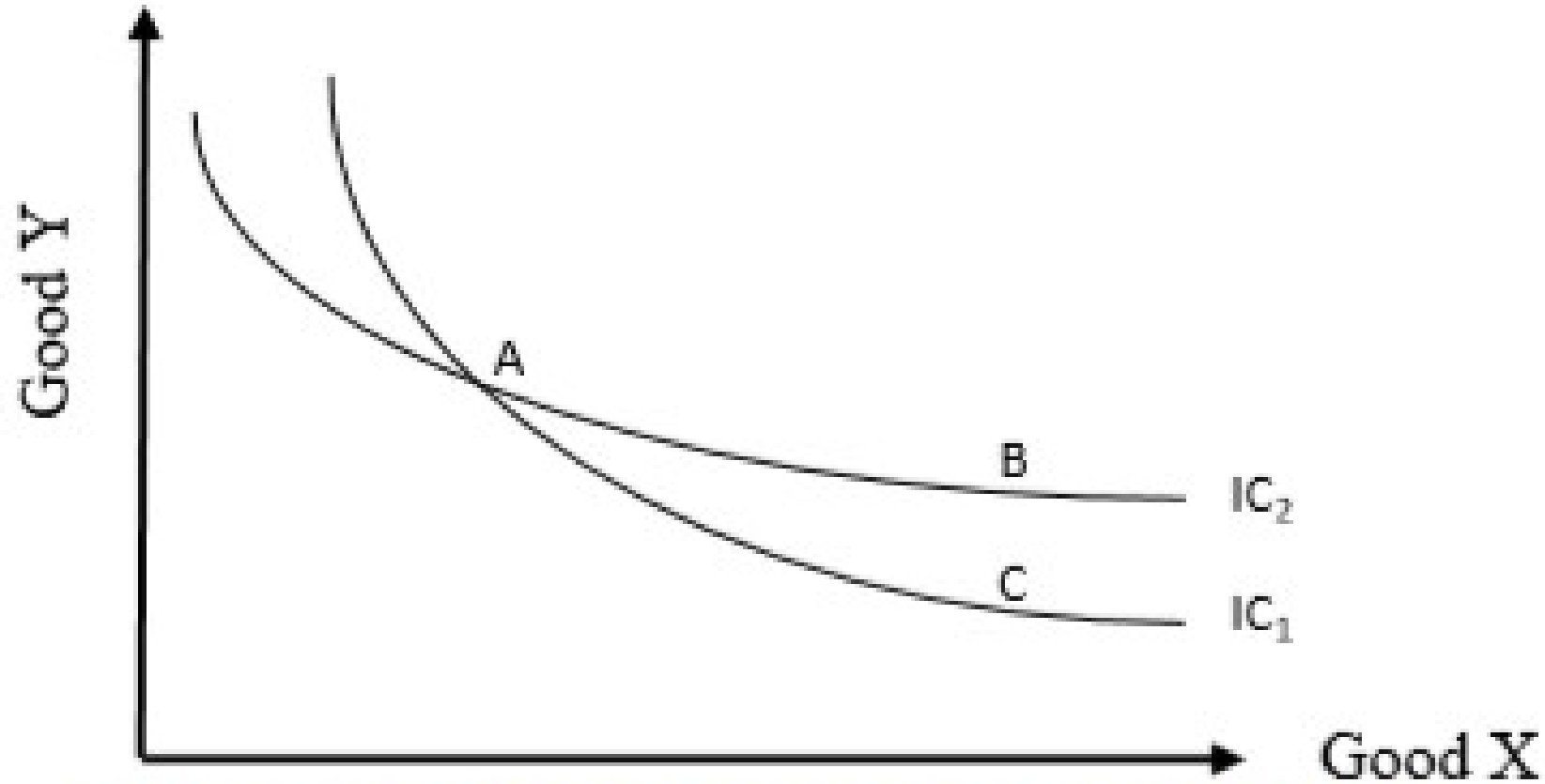


Figure 3.4: Intersection of indifference curves

Note : The intersection of indifference curves the same bundles of commodities give different utility.

3.3.2.4 Marginal rate of substitution (MRS)

- ❖ **Marginal rate of substitution** is a rate at which consumers are **willing to substitute one commodity for another** in such a way that the consumer remains on the same indifference curve.
- ❖ It shows **a consumer's willingness to substitute one good for another** while he/she is indifferent between the bundles.
- ❖ **MRS of X for Y ($MRS_{X,Y}$)** is defined as the number of units of commodity **Y that must be given up** in exchange **for an extra unit of commodity X** so that the consumer maintains the same level of satisfaction.
- ❖ Since one of the goods is sacrificed to obtain more of the other good, the **MRS is negative**.
- ❖ Hence, usually we take the absolute value of the slope.

$$MRS_{XY} = \frac{\text{Number of units of Y given up}}{\text{Number of units of X gained}} = \frac{\Delta Y}{\Delta X}$$

Let's consider the following indifference curve

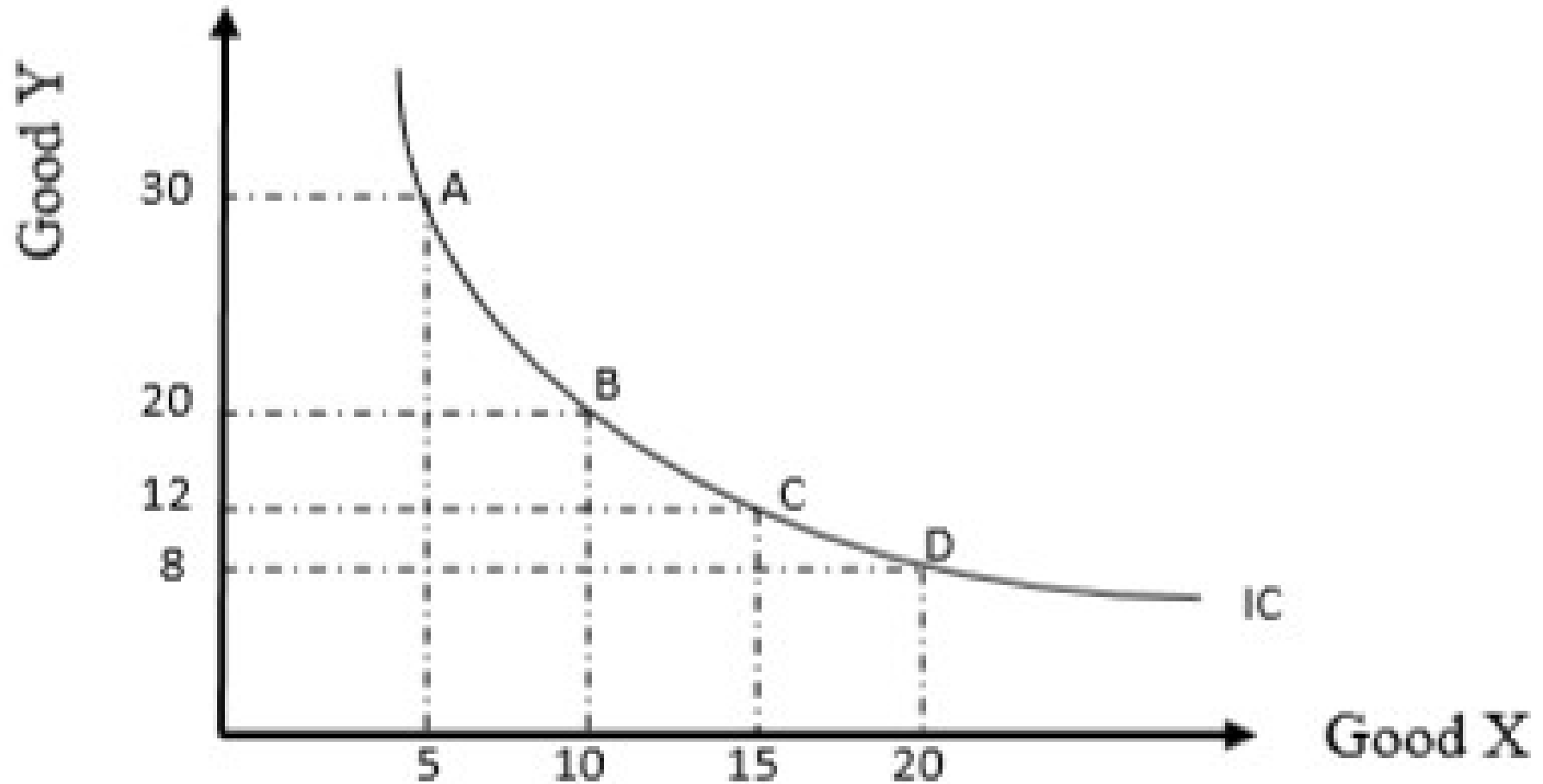


Figure 3.5: Indifference curve for two products X and Y

- ❖ From the above graph, $MRS_{X,Y}$ associated with the movement from point **A** to **B**, point **B** to **C** and point **C** to **D** is **2.0**, **1.6**, and **0.8** respectively.
- ❖ That is, for the same increase in the consumption of good X, the amount of good Y the consumer is willing to sacrifice diminishes.
- ❖ This **principle of marginal rate of substitution** is reflected by the **convex shape of the indifference curve** and is called **diminishing marginal rate of substitution**.
- ❖ It is also possible to derive MRS using the **concept of marginal utility**.
- ❖ $MRS_{X,Y}$, is related to MU_X and MU_Y as follows:

$$MRS_{X,Y} = \frac{MU_x}{MU_y}$$

Proof: Suppose the **utility function for two commodities** X and Y is defined as:

$$U = f(X, Y)$$

❖ Since **utility is constant along an indifference curve**, the **total differential of the utility function will be zero**.

$$dU = \frac{\partial U}{\partial X} dX + \frac{\partial U}{\partial Y} dY = 0$$

$$MU_X dX + MU_Y dY = 0$$

$$\frac{MU_X}{MU_Y} = -\frac{dY}{dX} = MRS_{X,Y}$$

Example: Suppose a consumer's utility function is given by $U(X, Y) = X^4 Y^2$
Find $MRS_{X,Y}$.

Solution: $MRS_{X,Y} = \frac{MU_X}{MU_Y}$; $MU_X = \frac{\partial U}{\partial X} = 4X^3 Y^2$ and $\frac{\partial U}{\partial Y} = 2X^4 Y$

$$MRS_{X,Y} = \frac{4X^3 Y^2}{2X^4 Y} = 2 \frac{Y}{X}$$

3.3.2.5 The budget line or the price line

- ❖ Indifference curves only tell us about consumer preferences for any two goods but they cannot show which combinations of the two goods will be bought.
- ❖ In reality, the consumer is constrained by his/her income and prices of the two commodities.
- ❖ This constraint is often presented with the help of the budget line.
- ❖ The budget line is a set of the commodity bundles that can be purchased if the entire income is spent.
- ❖ It is a graph which shows the various combinations of two goods that a consumer can purchase given his/her limited income and the prices of the two goods.

- ❖ In order to draw a budget line facing a consumer, we consider the following assumptions.
 - ✓ There are **only two goods** bought in quantities, say, **X** and **Y**.
 - ✓ Each consumer is confronted with **market determined prices**, **P_x** and **P_y** .
 - ✓ The consumer has a **known and fixed money income (M)**.
- ❖ Assuming that the consumer spends all his/her income on the two goods (X and Y), we can express the budget constraint as:

$$P_x Q_x + P_y Q_y = M$$

- ❖ By rearranging the above equation, we can derive the following general equation of a budget line:

$$Q_y = \frac{M}{P_y} - \frac{P_x}{P_y} Q_x$$

Graphically,

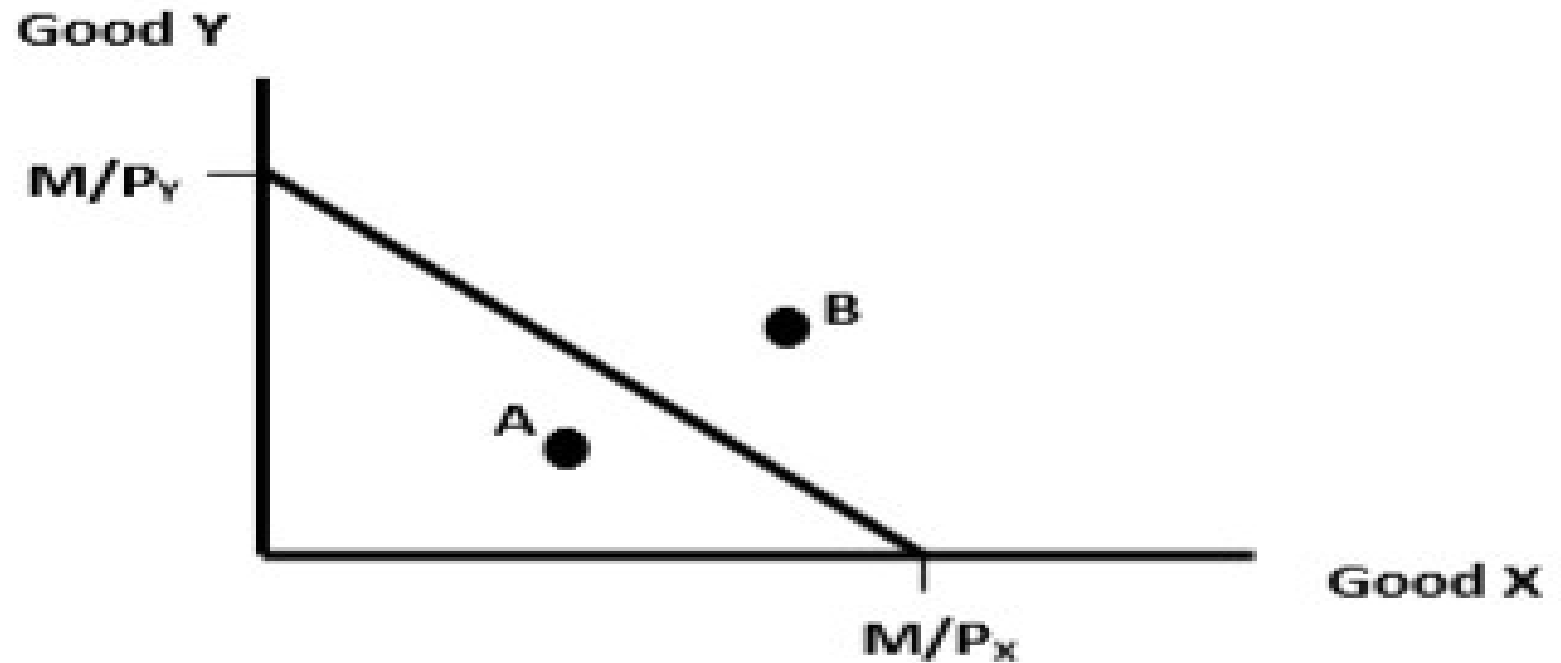


Figure 3.6: The budget line

Note:

- ✓ The slope of the budget line is given by $-\frac{P_X}{P_Y}$
- ✓ Any combination of the two goods within the budget line (such as point A) or **along the budget** line is **attainable**.
- ✓ Any combination of the two goods outside the budget line (such as point B) is unattainable (**unaffordable**).

Example: A consumer has \$100 to spend on two goods X and Y with prices \$3 and \$5 respectively. Derive the budget line equation and sketch the graph.

Solution:

$$P_x Q_x + P_y Q_y = M$$

$$3Q_x + 5Q_y = 100$$

$$5Q_y = 100 - 3Q_x$$

$$Q_y = \frac{100}{5} - \frac{3}{5}Q_x ; Q_y = 20 - \frac{3}{5}Q_x$$



Note: A budget is drawn for **given prices** and **fixed consumer's income**. Hence, the changes in prices or income will affect the budget line.

- ❖ **Change in income:** If the income of the consumer changes (keeping the prices of the commodities unchanged), the budget line also shifts (changes).
- ❖ **Increase in income causes** an **upward/outward shift** in the budget line that allows the consumer to buy more goods and services and,
- ❖ **decreases in income** causes in the budget line that leads the consumer to buy less quantity of **a downward/inward shift** he two goods.
- ❖ It is important to note that the **slope of the budget line** (the ratio of the two prices) does not change when income rises or falls.

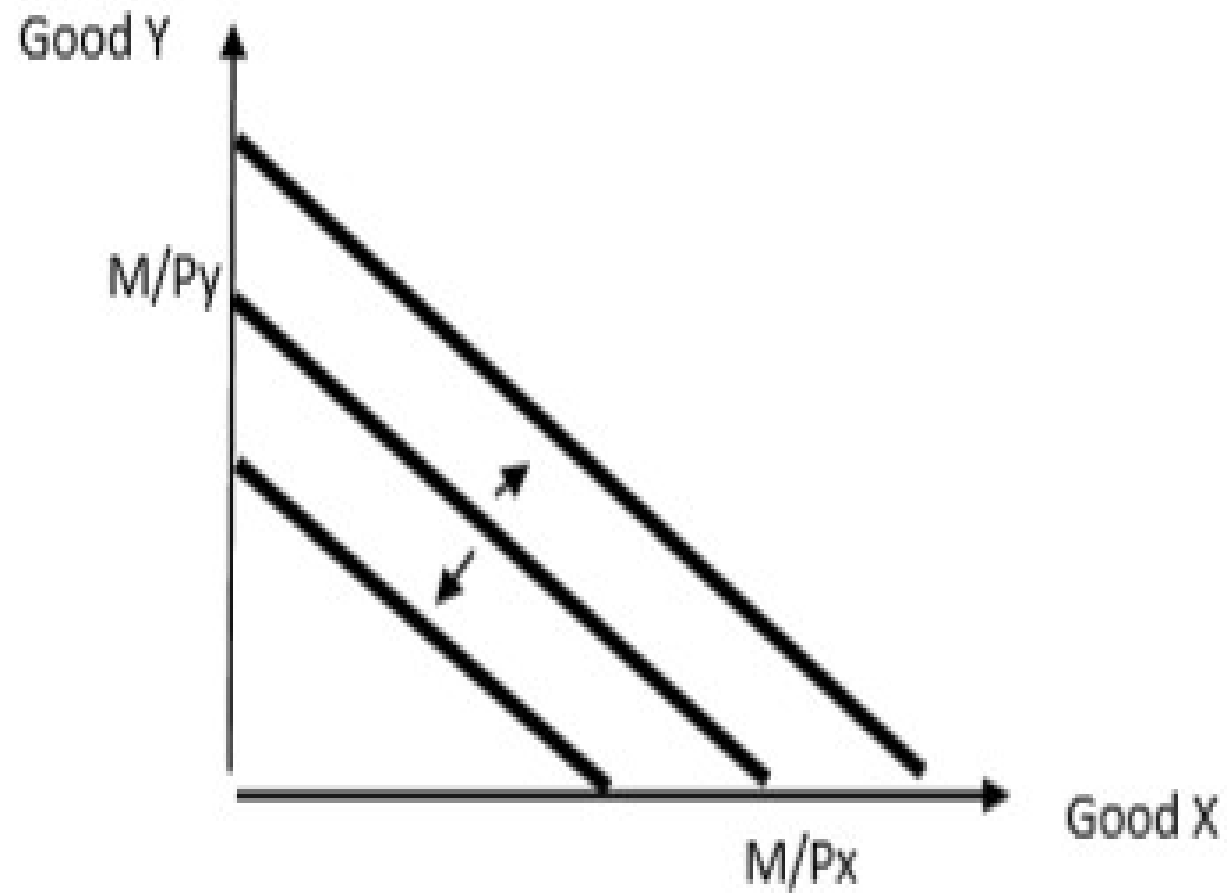


Figure 3.7: Effects of increase (right) and decrease (left) in income on the budget line

Change in prices:

- ❖ An **equal increase(proportion)** in the prices of the two goods **shifts the budget line inward**. Since the two goods become expensive, the consumer can purchase the lesser amount of the two goods.
- ❖ An **equal decrease(proportion)** in the prices of the two goods, on the other hand, shifts the **budget line out ward**. Since the two goods become cheaper, the consumer can purchase the more amounts of the two goods.

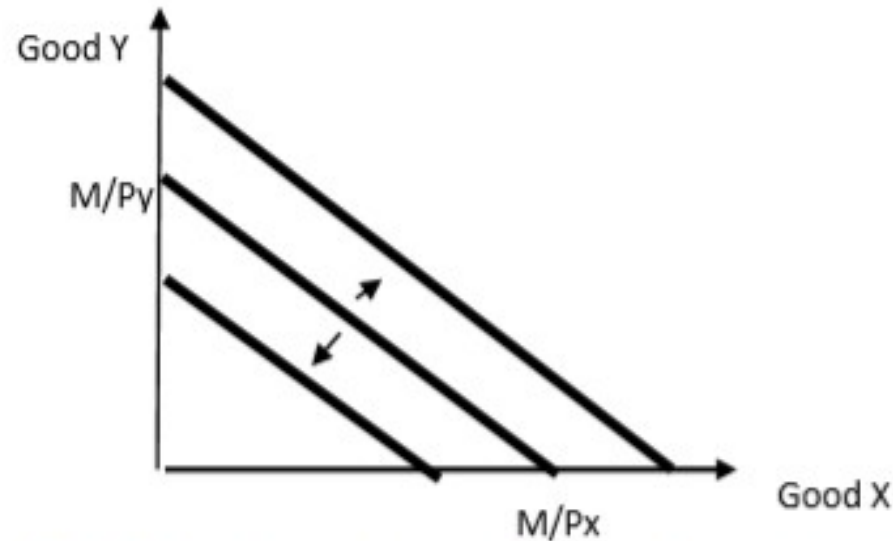


Figure 3.8: Effect of proportionate increase (inward) and decrease (out ward) in the prices of both goods

- ❖ **An increase or decrease** in the price of one of the two goods, keeping the price of the other good and income constant, **changes the slope of the budget line** by affecting only the intercept of the commodity that records the change in the price.
- ❖ **For instance**, if the price of good X decreases while both the price of good Y and consumer's income remain unchanged, the horizontal intercept moves outward and makes the budget line flatter. The reverse is true if the price of good X increases.
- ❖ On the other hand, if the **price of good Y decreases** while both the price of good X and consumer's income remain unchanged, the vertical intercept moves upward and makes the **budget line steeper**. The reverse is true for an increase in the price of good Y.

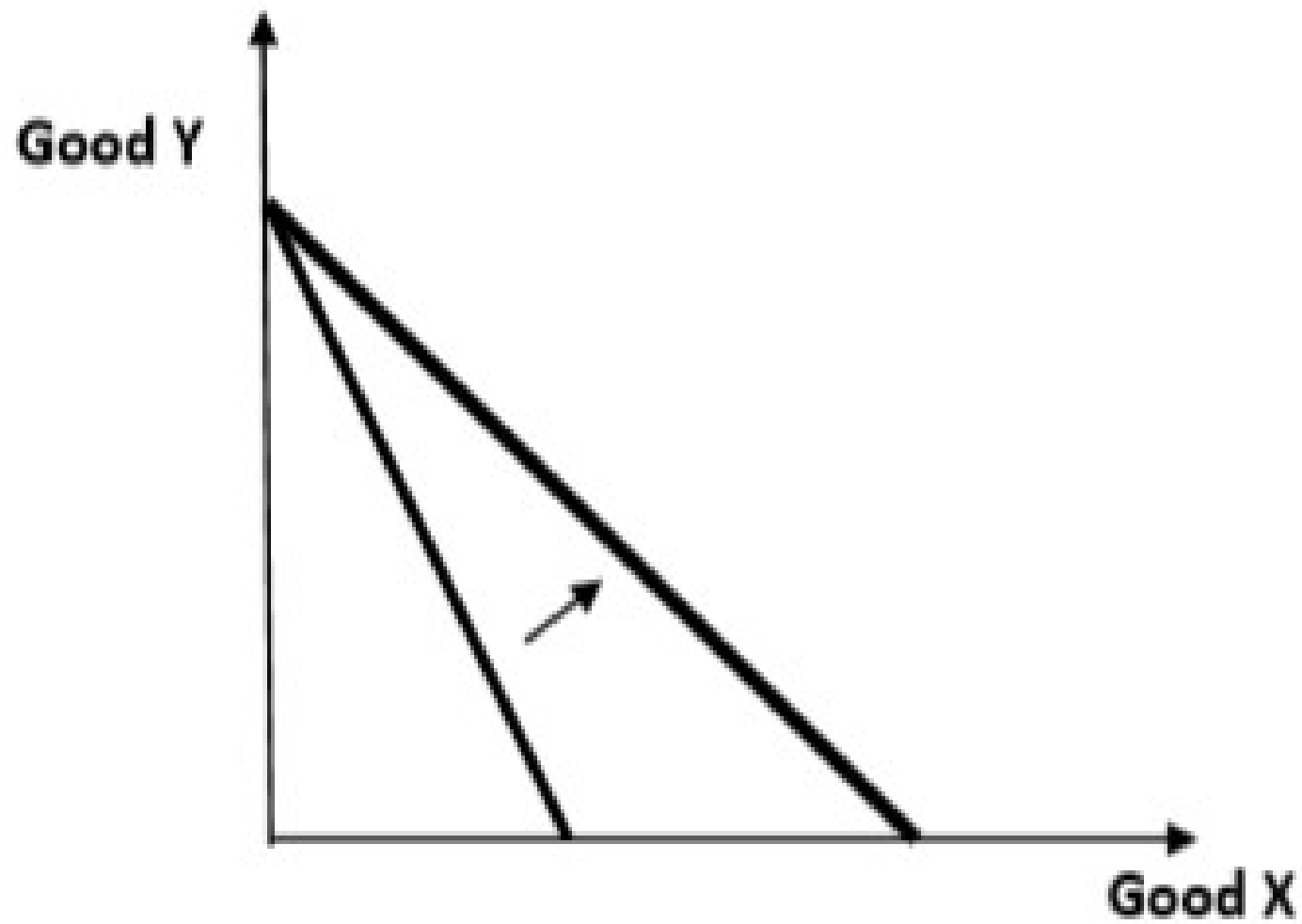


Figure 3.9: Effect of decrease in the price of only good X on the budget line

3.3.2.6 Equilibrium of the consumer

- ❖ The **preferences of a consumer** (what he/she wishes to purchase) are indicated **by the indifference curve**.
- ❖ The **budget line** specifies different combinations of two goods (say X and Y) the **consumer can purchase** with the limited income.
- ❖ **Therefore**, a rational consumer tries to attain the highest possible indifference curve, given the budget line.
- ❖ This occurs at the point where **the indifference curve is tangent** to the budget line so that the **slope of the indifference curve** ($MRS_{X,Y}$) is equal to **the slope of the budget line** ($-\frac{P_X}{P_Y}$).

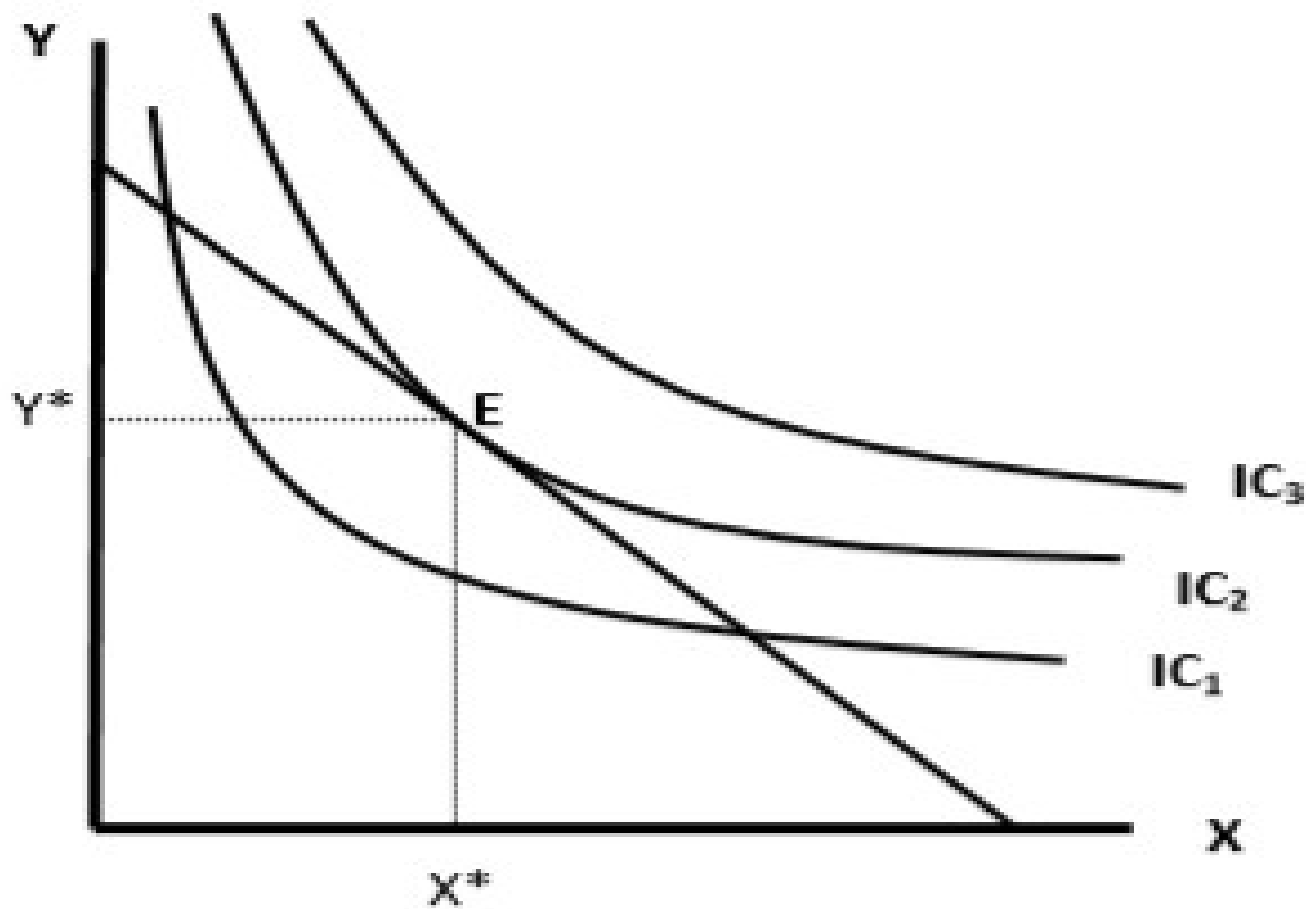


Figure 3.10: Consumer equilibrium under indifference curve approach

- ❖ In figure 3.10, **the equilibrium** of the consumer is at **point E**, where the budget line is tangent to the highest attainable indifference curve (IC_2).

❖ **Mathematically**, consumer optimum (equilibrium) is attained at the point where:

Slope of indifference curve = Slope of the budget line

$$MRS_{X,Y} = - \frac{P_X}{P_Y}$$

→ $\frac{MU_X}{MU_Y} = - \frac{P_X}{P_Y}$

Example: A consumer consuming two commodities X and Y has the utility function $U(X, Y) = XY + 2X$. The prices of the two commodities are 4 birr and 2 birr respectively. The consumer has a total income of 60 birr to be spent on the two goods.

- a) Find the utility maximizing quantities of good X and Y.
- b) Find $MRS_{X,Y}$ at equilibrium.

Solution:

a) The budget constraint of the consumer is given by:

$$4Q_x + 2Q_y = 60$$

$$Q_y = \frac{60}{2} - \frac{4}{2}Q_x ; Q_y = 30 - 2Q_x \dots\dots\dots \text{Budget Line}$$

✓ Thus, the slope of the budget line = 2

✓ **Slope of the IC** = $MRS_{X,Y} = \frac{MU_X}{MU_Y} = \frac{Y+2}{X}$

✓ **At Equilibrium:** **Slope of the IC** = slope of the budget line

$$\frac{Y+2}{X} = 2; Y+2 = 2X ; \qquad Y = 2X$$

✓ Substituting this in the budget equation results

$$Y = 14 \text{ and } X = 8$$

$$\text{b) } MRS_{X,Y} = \frac{MU_X}{MU_Y} = \frac{Y+2}{X} = \frac{14+2}{8} = \frac{16}{8} = 2$$